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Physicochemical and tribological properties of microalgae oil as biolubricant for hydrogen-powered engine☆

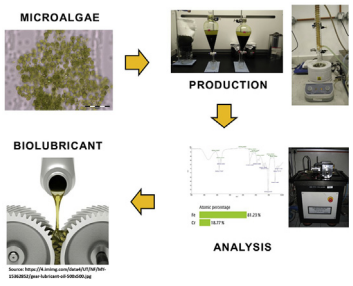
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HIGHLIGHTS

- Chemically modified *Chlorella* sp. microalgae oil is used to produce biolubricants.
- The coefficient of friction are reduced up to 10.1% with modified microalgae oil additions.
- Significant wear loss and surface roughness reductions on modified microalgae oil blend-lubricated surfaces.
- Lubricant's heat dissipation are also enhanced with modified microalgae oil additions.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrogen fuel offers a cleaner fuel alternative to fossil fuel due to more efficient burning as well as reduces the environmental and health issues brought by fossil fuel usage. In engine application, regardless of either pure hydrogen or in combination with air or/and other biofuel, all the moving parts are exposed to friction and wear, and lubricant is used to minimize friction and wear for optimum operation. Thus, in this study, the use of microalgae oil as an alternative biolubricant is evaluated from the physicochemical and tribological aspects. It is found that modified microalgae oil (MMO) has demonstrated great

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anti-friction and anti-wear potential, particularly the 10% modified microalgae oil blend (MMO-10). The coefficient of friction is reduced (up to 10.1%) and significant reductions of wear loss and surface roughness are obtained in comparison to pure poly-alpha-olefin. Lubricant's heat dissipation is also enhanced with MMO addition, demonstrating great prospect for MMO for hydrogen-powered engine utilization.

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Introduction

Renewable energy has offered a solution for global warming due to greenhouse gas emissions from the burning of fossil fuels. Among the types of renewable energy widely implemented in countries around the world are solar, wind, wave, bioenergy, biofuel and hydro [1–4]. However, one major problem associated with renewable energies like solar and wind is the intermittent availability of the energy source, thus the need for storage device. Yet, the only storage device commercially available in the market is the battery. This has led researchers to look for better performing materials to increase the efficiency for energy storage such as phase change materials and graphene [5,6]. But one potential renewable energy that can be used like fuel is hydrogen energy. This is one of the reasons why the use of hydrogen in internal combustion engines has gained tremendous attention worldwide [7], as it provides a possible solution to fossil fuel depletion. Hydrogen fuel offers a cleaner fuel alternative due to more efficient burning, thus reduces the negative environmental impact like greenhouse gas emissions and health issues brought by fossil fuel usage [8]. However, regardless of how hydrogen fuel is used in an engine, i.e. pure hydrogen or in combination with air or/and other biofuel [9–11] etc., the moving components are exposed to friction and wear, and lubricant is introduced to reduce the friction and wear generated.

The global lubricant market was valued at USD 118.89 billion in 2016, and it is forecasted to reach 45.4 million metric tons in 2019, with a projected rise of 2% per year [12]. Since conventional lubricant are mainly produced from petroleum, biolubricants have received considerable attention from the scientific community, as they are renewable, biodegradable and non-toxic [13–15], providing a potential solution to address the health and environmental issues facing the petroleum-based counterpart. Moreover, vegetable oil is used for biolubricant production [16], which is greatly affected by the fluctuating commodity prices [17] and the food versus fuel dilemma. Due to these concerns, scientists start to develop biofuel from non-edible oil sources, and among the widely studied are rice bran [18], *Calophyllum inophyllum* [19], *Ceiba pentandra* [20], jatropha [21] and waste cooking oil [22], collectively known as second-generation biofuel. And past research have also revealed that these biofuels offered better lubricity than pure diesel. However, the conflict of arable land still persist since both the first- and second-generation biofuel feedstock still require a substantial area for cultivation. Thus, microalgae marked as the third-generation biofuel has resolved the arable land competition issue, since they can be cultivated in photobioreactors on marginal land [23]. Both microalgae-based biodiesel [24–29] and microalgae-based bio-oil [30–32] are believed to demonstrate exceptional lubrication properties just like their vegetable-oil counterparts. Likewise, as with many biodiesels, microalgae-based oil can be reintroduced as biolubricant. Past studies have revealed that microalgae bio-oil offers friction and wear benefits but crude bio-oil cannot be used directly due to the high acidity, and the properties of crude bio-oil can be enhanced through different technologies such as emulsification [33], catalytic fluid cracking [34], hydrogenation [35], catalytic esterification [36] and distillation [37]. Among these methods, catalytic esterification is found to be the most effective in corrosiveness reduction in bio-oil [32]. Even though the upgrading of microalgae bio-oil's properties are being widely research, the literatures related to its tribological characteristics are still minimal.

Xu et al. [36] evaluated the tribological performance of methyl-esterified bio-oil from the microalgae *Spirulina*. Crude bio-oil were upgraded chemically via the esterification process to reduce the corrosiveness of the bio-oil and carboxylic acids have been identified as the cause for corrosion and materials removal in the frictional pairs. They found that esterified bio-oil showed better friction and wear than the crude bio-oil since the esterification reaction converted the carboxylic acids into esters, which increases the ester content and reduces the acidity accordingly [38]. The presence of a tribofilm and the adsorbed oil film have served as a protective layer between the frictional pairs, reducing the friction and wear. Also, the higher amount of ester content resulted in better lubrication properties due to the formation of a thicker lubricating film. In another study, it was revealed that *Spirulina* bio-oil esterified using catalyst with stronger alkalinity was better performing frictionally than bio-oil esterified with a less alkaline catalyst, even though both the esterified bio-oils still performed better than crude bio-oil [14]. The results showed that the friction coefficients of esterified bio-oil were reduced by 22% (with more alkaline catalyst) and 10% (with less alkaline catalyst) in comparison to crude bio-oil. The improvement of friction observed was attributed to the more robust lubricating film formed by the upgraded oils due to stronger adsorption of the ester groups on metal surfaces in addition to the existence of a tribofilm of Fe_2O_3 on the metal surfaces.

In this study, the friction and wear characteristics of microalgae-based biolubricants for hydrogen-powered engine applications are presented. First, crude microalgae oil was

derived from dried *Chlorella* biomass. The crude oil was upgraded using catalytic esterification to obtain the modified microalgae oil for quality improvement. Then, the tribological performance of the modified microalgae oil was evaluated using a high frequency reciprocating rig with pin-on-plate configuration, and the corresponding lubricating mechanism was elucidated.

Methodology

Formulation of microalgae-based biolubricants

Extraction of CMO

The modified Folch method [39] was used to extract the CMO from 500 g of *Chlorella* sp. biomass using a mixture of chloroform and methanol (2:1, v/v) for 12 h at 500 rpm. This was followed by filtration through a 1-μm polytetrafluoroethylene filter to remove biomass residues. The filtrate was washed with 1% sodium chloride solution at a volume ratio of 8:4:3 (chloroform: methanol: 1% sodium chloride solution). The mixture was shaken and left to settle in a separatory funnel for 12 h. Once the separation process was complete, the bottom layer (CMO) was collected and the excess solvents were evaporated from the CMO using a rotary evaporator (Model: RV 10 digital V, IKA®-Werke GmbH & Co. KG, Germany). The extracted CMO was weighed and its acid value was measured.

Acid-catalyzed esterification and alkaline-catalyzed transesterification

CMO cannot be used directly in engines and machinery due to its high acidity, which will lead to corrosion of the parts. Thus, acid-catalyzed esterification and alkaline-catalyzed transesterification were carried out in order to improve the physicochemical properties of the CMO. In acid-catalyzed esterification, the CMO was esterified with methanol (50%, v/v) using 1 (v/v) % of sulfuric acid as a catalyst. The reaction was carried out for 2 h at a temperature of 65 °C and stirring speed of 500 rpm. Once the esterification process was complete, the reaction mixture was left to settle overnight in a separatory funnel. Two distinct layers formed in the separatory funnel after the separation process, where the top layer was the esterified oil and the bottom layer was a mixture of glycerin and other impurities. The esterified oil was collected and transferred into a rotary evaporator in order to evaporate excess methanol and water.

Next, the esterified microalgae oil was transesterified with 50 (v/v)% of methanol using 1 (w/w)% of potassium hydroxide (KOH) as a catalyst for 2 h at a temperature of 65 °C and stirring speed of 500 rpm. Once the transesterification process was complete, the reaction mixture was left to settle overnight in a separatory funnel. Two distinguishable layers formed in the separatory funnel, where the top layer was the MMO and the bottom layer was a mixture of glycerin, excess of methanol and other impurities. The MMO was collected and transferred into a rotary evaporator in order to evaporate excess methanol. The weight and acid value of the MMO were measured.

Preparation of the MMO blends

PAO (Durasyn® Grade 170, INEOS Oligomers LLC, USA) was used as the base oil for the MMO blends. Five samples were prepared by blending the PAO with 1, 2, 5, 10, and 20 (v/v)% of MMO.

Fatty acid methyl ester content and physicochemical properties of the biolubricants

Fourier transform infrared spectroscopy, gas chromatography, and gas chromatography-mass spectrometry analyses

Fourier transform infrared (FTIR) spectrometer (Model: Spectrum 400, PerkinElmer Instruments LLC, USA) was used to analyze the compositions of the CMO and MMO. The absorption spectra were recorded within a wavenumber range of 4000–400 cm^{-1}. The fatty acid methyl ester (FAME) content of the MMO was determined according to EN 14103 using a gas chromatograph with flame ionization detector (GC-FID) (Model: 7890A, Agilent Technologies Inc., USA). Agilent HP-INNOWax column (length × inner diameter × film thickness: 30 m × 0.25 mm × 0.25 μm) was chosen as the capillary column for the GC-FID analysis. The oven temperature was programmed according to the following sequence: (1) hold at 60 °C for 2 min, (2) heat from 60 to 200 °C at 10 °C/min, (3) heat from 200 to 240 °C at 5 °C/min, and (4) hold at 240 °C for 7 min. Nitrogen was used as the carrier gas. The flame ionization detector was set at a temperature of 250 °C, the flow rate was set as 2 mL/min, the type of injector was splitless, and the injection volume was 1 μL. The FAME content of the samples was compared with a certified FAME standard mixture (Supelco® 37 component FAME mix, Supelco Inc., USA). The FAME content (ℂ) was calculated using the following equation: $\mathbb{C} = \frac{\sum A - A_{EI}}{A_{EI}} \times \frac{W_{EI}}{W} \times 100$, where $\sum A$ is the total peak area from the methyl ester in C4:0 to that in C22:6; A_{EI} is the peak area corresponding to nonadecanoic acid methyl ester; W_{EI} is the weight, in milligrams, of the nonadecanoic acid methyl ester being used as internal standard and W is the weight, in milligrams, of the sample.

The compounds present in the MMO were analyzed using gas chromatograph-mass spectrometer (GC-MS) (Model: GCMS-QP2010 Ultra, Shimadzu Corporation, Japan) with a low-bleed column (Model: Rtx-5MS, Restek Corporation, USA, length × inner diameter × film thickness: 30 m × 0.25 mm × 0.25 μm). The injection volume was 1 μL. The oven temperature was programmed according to the following sequence: (1) hold at 60 °C for 2 min, (2) heat from 60 to 200 °C at 10 °C/min, (3) heat from 200 to 250 °C at 5 °C/min, and (4) hold at 250 °C for 7 min. Helium was used as the carrier gas with a split injection ratio of 100:1. The compounds (based on the peak area) were identified from the NIST 02 mass spectral library. The quadrupole mass spectrometer was operated at an interface temperature of 250 °C and the ion source temperature of 200 °C with a sample flow rate of 1 mL/min.

Physicochemical properties

The important physicochemical properties such as heating value, acid value, kinematic viscosity, dynamic viscosity, density at 15 °C, thermal stability and ester content of CMO,

MMO, MMO blends, PAO and FFL were measured according to EN and ASTM standard methods. Table 1 shows the list of equipment used in this study.

Thermogravimetric analysis was performed using a TGA (Model: TGA 4000, Perkin Elmer, USA). Sample of about 2–50 mg was heated at a constant heating rate of 15 °C/min from 30 °C to 700 °C in an inert environment. Nitrogen gas with a flowrate of 20 mL/min was used. Sample's weight loss against time was plotted to determine the decomposition temperature, the extrapolated onset temperature that denotes the temperature at which the weight loss begins.

Friction and wear characteristics

The friction and wear tests were conducted using a high frequency reciprocating rig (Model: TR-281-M8, Ducom Instruments, USA) with pin-on-plate configuration. ASSAB DF2 carbon steel plate (length × width × thickness: 15 mm × 15 mm × 4 mm) was chosen as the test plate whereas a cylindrical roller bearing (Part no.: RC-6X6 C, SKF, Sweden) was chosen as the pin. Fresh test plate and pin were used for each sample. The COF values were recorded automatically during the tests. The test conditions of the high frequency reciprocating rig are presented in Table 2. PAO synthetic lubricant and fully formulated synthetic lubricant (FFL) was used as the control samples. The test plates were washed with acetone and dried in the air prior to the surface morphology and elemental analysis using scanning electron microscope (SEM) integrated with an energy-dispersive X-ray spectrometer (EDS) (Model: Phenom ProX, Phenom-World B.V., The Netherlands). Surface roughness studies of wear tracks were conducted using a profilometer (Model: Infinite Focus, Alicona Imaging GmbH, Austria).

Results and discussion

Physicochemical properties of CMO and MMO

The physicochemical properties of CMO and MMO are presented in Table 3. The physical appearance of the CMO and MMO is almost grease-like. The CMO yield (CMO extracted from the dry biomass) is 70% whereas the MMO yield is 83%, indicating the yield is significantly increased after chemical modification where free fatty acids were successfully converted into an ester. Our results further revealed that the total ester (FAME) content in MMO is not that high, only 48.48 (m/m)%, indicating that other fatty compounds were also present as shown in the GC-MS analysis (Table 4). This is due to microalgae only produce very small amount of neutral lipid (triglyceride) under optimum growth that can be converted into fatty acid methyl ester.

Based on the GC-MS analysis, the MMO is composed of esters, sterols, ketones, peroxides, and ether, as shown in Table 4. Cellulose residues such as furan derivatives are present, Xu et al. [14] also found that furan derivatives and amino acids were present in the biooil derived from *Spirulina* sp.. Phytol (which is diterpene alcohol from chlorophyll) as well as plant sterols such as stigmasterol and β-sitosterol were also present. Phytol is widely used in the cosmetics and food

Table 1 – List of equipment and standard methods used for physicochemical properties characterization.

Properties	Equipment	Model and manufacturer	Standard method	Accuracy
FAME content	Gas chromatograph with flame ionization detector (GC-FID)	7890A, Agilent Technologies Inc., USA	EN 14103	±0.008%
Heating value	Bomb calorimeter	6100, Parr Instrument Company, USA	ASTM D240	±0.001 MJ/kg
Acid value	Potentiometric titrator	G20, Mettler Toledo, USA	ASTM D664	±0.001 mgKOH/g
Kinematic viscosity	Stabinger viscometer	SVM 3000TM G2, Anton Paar GmBH, Austria	ASTM D7042	±0.001 mm^2/s
Dynamic viscosity	Stabinger viscometer	SVM 3000TM G2, Anton Paar GmBH, Austria	ASTM D7042	±0.001 mPas
Density at 15 °C	Stabinger viscometer	SVM 3000TM G2, Anton Paar GmBH, Austria	ASTM D4052	±0.1 kg/m^3
Thermal stability	Thermogravimetric analysis (TGA)	TGA 4000, Perkin Elmer, USA	ASTM E2550	±1.00 °C

Table 2 – Test conditions for the high frequency reciprocating rig.

Test condition	Value
Reciprocating frequency	10 Hz
Temperature	75 °C
Duration	60 min
Load	100 N

Table 3 – Physicochemical properties of the CMO and MMO.

Properties	Test method	Unit	CMO	MMO
Acid value	ASTM D664	mg KOH/g	49.688	19.703
Heating value	ASTM D240	MJ/kg	27.078	32.939
Total ester content	EN 14103	(m/m)%	n.a.	48.48

Note: n.a. denotes "not available".

industries because of its aromatic properties while stigmasterol and β-sitosterol are lucrative secondary metabolites which are marketed as dietary supplements in the nutraceutical industry due to their antioxidant and anti-inflammatory properties [40,41]. These oxygen-containing compounds and polyaromatics are natural lubricity enhancers, as indicated by the results of previous studies [42,43]. The ability of these polar functional groups to form a protective layer on the metal surfaces minimizes wear by preventing direct metal-to-metal contact when sliding against each other. Furthermore, the arrangement of fatty acid on the matting surfaces by the adsorption of the polar carboxyl head on the solid surface, leaving the alkyl non-polar tail facing the opposite surface will create a molecular barrier between the rubbing surfaces [44].

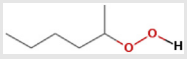
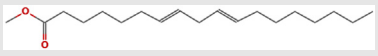
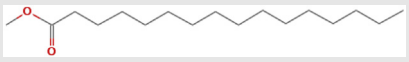
For biofuel or biolubricant production purposes, low neutral lipid content is not desirable but environmental and nutrient stresses can be employed to induce lipid accumulation. The microalgae cells will only start to accumulate lipid as energy and carbon storage [45] when subjected to environmental stresses such as nutrient starvation [46] and high irradiance [47,48]. During these periods, other cell functions like protein synthesis and cell division will decrease. Research has found that for some microalgae species, all available carbon including starch from chlorophyll will be converted into lipid when they are subjected to nitrogen-deplete condition for an extended period [49]. Chlorophyll will be disintegrated during this period to save energy and the newly made lipid bodies will now occupy the space previously used to store the plastid and chlorophyll. Thus, for optimum lipid production, it is proposed that microalgae are first be cultured in nutrient rich media to attain the desirable cell numbers before any lipid inducers are introduced [50], in order not to compromise the overall biomass and lipid productivity.

The lipid composition and chain length of MMO are presented in Fig. 1. The total ester content is primarily composed of polyunsaturated fatty acid methyl esters (68.7%), whereas monounsaturated and saturated fatty acid methyl esters constitute 19.1 and 12.2%, respectively. Geller and Goodrum [51] found that lubricity is improved as unsaturation increases. In this study, MMO contained 87.8% of total unsaturated fatty acid methyl esters. On the contrary, other researcher has found that unsaturation will give rise to isomer, and isomer especially the *cis*-configuration will prevent the close packing of fatty acids of the formed boundary lubrication film [52] thus reduces the film effectiveness. Despite of the fact that the degree of unsaturation may affect the effectiveness of the formed film, another important factor to consider is the carbon chain length of the fatty acids [51]. Based on the results, it is found that MMO composed of only long chain fatty acids, LCFA (13–21 carbons) and very long chain fatty acids, VLCFA (>21 carbons) at 63 and 37% respectively. Long carbon chain are able to create a thicker film compared with the short fatty acid chains. Talib and Rahim [53], also arrived at the same conclusion when modified jatropha oil (MJO) that contained C16 to C18 was compared with the synthetic ester (SE) (containing C8 to C10), the friction coefficient recorded for MJO was lower than SE. Furthermore, the strong molecular barrier is formed due to the close packing of straight long-chain fatty acids, on top of the lateral inter-molecule interactions between the carbon tails that will further strengthen the structure of the established-film [54]. A strong and thick molecular film reduces wear [55].

It can be seen from Table 3 that the CMO has a very high acid value (49.688 mg KOH/g), which was not a desirable for properties for lubricant as it will cause corrosion. High free fatty acid content can cause severe corrosion to engine parts [56] besides affecting the storage stability of ester-based lubricant. Water, air or acidic condition exposures will accelerate the initiation of oxidation [57], that alter the esters' chemical composition. Thus, a lubricant's storage stability is indirectly prolonged by lowering the acid value. In this study, it is found that the acid value is significantly reduced to 19.703 mg KOH/g after the chemical modification, which corresponds to more than 50% reduction of CMO's acidity. The current acid value is still relatively high since the esterification reaction was not optimized. The used of a higher esterification temperature might be effective in enhancing the efficiency of esterification reaction, but a higher reaction temperature may also cause the oxidation of the MMO's esters, which degrades the MMO oil and compromises its quality. In addition, the higher temperature will cause the methanol to evaporate, reducing the overall efficiency of the reaction. The used of a catalyst with higher alkalinity may help in the further reduction of MMO's acidity, where a lower acid value was recorded by the microalgae bio-oil sample esterified with methanol using a higher alkalinity catalyst compared with a lesser alkalinity ones [36].

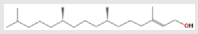
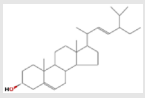
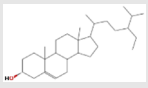
In addition, the heating value of the MMO (32.939 MJ/kg) is higher than that for the CMO (27.078 MJ/kg), which indicates that the MMO has higher energy content. Hong et al. [58] found that the heating value for fatty acid alkyl esters increases with the increase of molecular weight but decreases when the number of double bonds increases. In this study, the lubricant's molecular weight increases when a hydrogen (H) from carboxylic acid was replaced with a methyl group (CH_3) during acid-catalyzed esterification. It can be seen from Fig. 2 that the peak at 1059 cm^{-1} (CMO) assigns to C–O stretching has shifted to 1025 cm^{-1} (MMO) after the chemical reaction. The peak intensity has increased from 0.20A to 0.34A due to the increase of C–O bond from the R group during esterification when H

Table 4 – Main compounds present in the MMO identified from GC-MS analysis.

No.	Compound	Molecular formula	Chemical structure	Retention time (min)	Peak Area (%)
1	Tetrahydrofuran, 2,2-dimethyl-	$C_6H_{12}O$		3.771	3.433
2	4-Butoxy-2-butanone	$C_8H_{16}O_2$		4.069	9.328
3	Hydroperoxide, 1-methylhexyl	$C_7H_{16}O_2$		4.341	0.219
4	Hydroperoxide, 1-ethylbutyl	$C_6H_{14}O_2$		4.446	2.465
5	Hydroperoxide, 1-methylpenthyl	$C_6H_{14}O_2$		4.612	3.385
6	Cyclopentane, 1-acetyl-1,2-epoxy-	$C_7H_{10}O_2$		4.843	11.873
7	Cyclopropane, 1,1,2,3-tetramethyl-	C_7H_{14}		5.300	0.573
8	Cyclopropane, 1,1,2,2-tetramethyl-	C_7H_{14}		5.482	0.173
9	7,10-Hexadecadienoic acid, methyl ester	$C_{19}H_{34}O_2$		17.969	0.406
10	Hexadecanoic acid, methyl ester	$C_{17}H_{34}O_2$		18.348	0.742
11	1-Docosanol, methyl ether	$C_{23}H_{48}O$		19.702	0.168
12	9,12-Octadecadienoic acid, methyl ester	$C_{19}H_{34}O_2$		20.693	1.094

(continued on next page)

Table 4 – (*continued*)

No.	Compound	Molecular formula	Chemical structure	Retention time (min)	Peak Area (%)
13	Phytol	$C_{20}H_{40}O$		20.951	0.167
14	Heneicosanoic acid, methyl ester	$C_{22}H_{44}O_2$		25.586	0.078
15	Stigmasterol	$C_{29}H_{48}O$		26.793	1.083
16	β-sitosterol	$C_{29}H_{50}O$		32.147	4.619

from the carboxylic acid group (–COOH) is substituted with carbon from CH_3, forming ester (-COOR). Lubricity increases due to the presence of more functional groups [5] available for metal-surface-adsorption, when more of the free fatty acids are converted into esters, creating a thicker molecular lubricating film.

The physicochemical properties of the PAO, FFL, and MMO blends are presented in Table 5. The density of MMO blends and FFL are almost similar but the dynamic viscosity and kinematic viscosity of MMO blends are generally lower than FFL. Meanwhile, the acid value increases with an increase in the percentage of MMO in the blend. A buffer solution containing a weak base and its conjugate acid can be applied to further reduce the acidity. Our analysis showed that all MMO blends except for MMO-20 fulfilled the ISO VG Grade 68 standard (Table 6). Some of the commercially available ISO VG Grade 68 gear and hydraulic oils are shown in Table 5. The slight difference in the density parameter was less crucial compared with the viscosity because the viscosity of an oil governs their lubrication application [59].

Thermal stability and thermal protection

Thermal stability is crucial for vegetable oil-based biolubricant application at high temperature condition [60] and the onset temperature is used to define the temperature at which a lubricant starts to decompose. The thermal stability of vegetable oil is predominantly affected by the chemical structure of its fatty acid composition [61]. Therefore, the fatty acid's properties like carbon chain length, branching, and

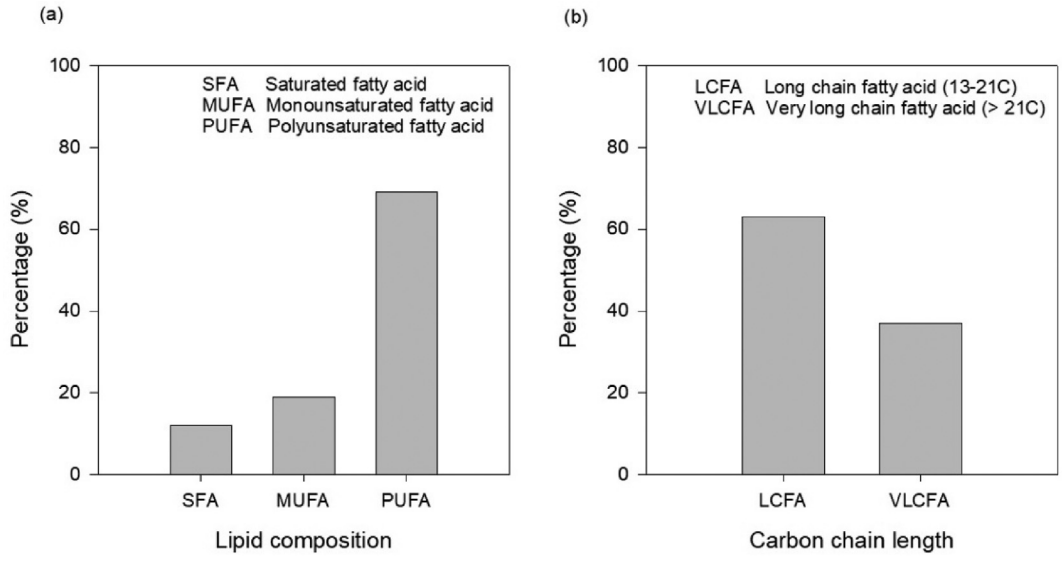


Fig. 1 – Lipid composition and carbon chain length of MMO.

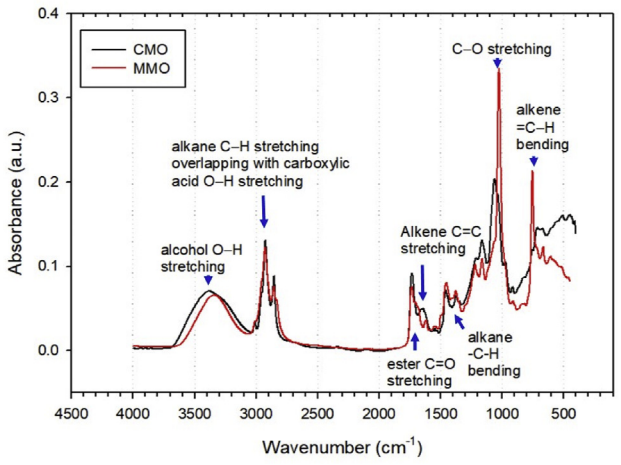


Fig. 2 – FTIR spectra of the CMO and MMO.

degree of unsaturation will directly influence the thermal stability of the vegetable oil [62]. Fig. 3 shows the thermal curves of PAO, MMO-10 and FFL, it is evident that MMO-10 has better thermal stability, of about 14 °C higher than FFL. The decomposition temperature of MMO-10 is at 333.33 °C compared with FFL at 318.98 °C. Similarly, Kalam et al. [60] also found that olive oil showed better thermal stability than a SAE15W40 commercial lubricant where the decomposition temperature of olive oil and the commercial lubricant were at 395.15 and 249.20 °C respectively. Furthermore, the thermal curve have shown that there were other compounds added to FFL, probably as additives. Higher decomposition temperature denotes higher functional temperature for the lubricant. When a lubricant decomposition temperature is exceeded, the temperature will increase dramatically as the fluid internal resistance to flow increases, causing an overheated condition. In this condition, the oil film would breakdown, losing its lubrication properties. This will lead to the increase of metal contacts between the rubbing surfaces, causing severe wear of components and the eventual system failure. Therefore, for biolubricant application, MMO-10 can be used at a slightly higher operating temperature than FFL, an added advantage from the higher thermal stability it possesses.

Nowadays, it is essential for lubricants to provide thermal protection against the heat generated during sliding motion, apart from providing wear protection. Fig. 4 shows the average temperature measured during the stable friction phase. The programmed test temperature was 75 °C and the results indicate that the MMO offers a certain degree of thermal protection. The average temperature of the MMO blends is lower by 0.19–0.37 °C compared with the average temperature for PAO. The average temperature of the MMO blends is lower by 0.14–0.32 °C compared with that for FFL. The results indicate that MMO improves heat dissipation of the lubricant, which is likely one of the reasons for the reduced wear on the MMO blend-lubricated surfaces as the progression of adhesive wear were reduced at lower temperatures, evidenced from the improved coefficient of friction and wear scar analysis. The oil viscosity has provided an effective thin lubrication film at the interacting interfaces that reduce the friction and heat generated [53] where on the overall, the viscosity for MMO blends were higher than that of PAO except for MMO-1. Hence,

Table 5 – Physicochemical properties of the PAO, FFL, and MMO blends.

Lubricant	Dynamic viscosity (mPa.s)	Kinematic viscosity (mm^2/s)	Density at 15 °C (g/cm^3)	Acid value (mg KOH/g)
Test method	ASTM D7042	ASTM D445	ASTM D4052	ASTM D664
PAO	54.2	66.2	0.835	0.179
MMO-1	54.1	66.0	0.835	0.938
MMO-2	59.7	72.8	0.836	1.082
MMO-5	59.0	71.5	0.841	3.624
MMO-10	57.9	69.4	0.850	5.019
MMO-20	66.2	78.6	0.857	13.688
FFL	64.2	76.7	0.852	0.258
Commercial ISO VG 68 lubricants				
Gear oil A	n.a.	68.0	1.079	n.a.
Gear oil B	n.a.	65.8	0.873	n.a.
Gear oil C	n.a.	65.0	0.870	n.a.
Hydraulic oil A	n.a.	53.1	0.877	n.a.
Hydraulic oil B	n.a.	68.0	0.897	n.a.
Hydraulic oil C	n.a.	68.0	0.880	n.a.
Hydraulic oil D	n.a.	68.0	0.880	n.a.

Note: "n.a." = not available; "A"; "B", "C", "D" = referred to different brands.

Table 6 – ISO VG grade 68 according to ISO 3448.

Parameter	Mid-point viscosity at 40 °C (mm^2/s)	Minimum viscosity at 40 °C (mm^2/s)	Maximum viscosity at 40 °C (mm^2/s)	Density at 15 °C (g/cm^3)
Test method	ASTM D445	ASTM D445	ASTM D445	ASTM D4052
ISO VG 68	68.0	61.2	74.8	0.865

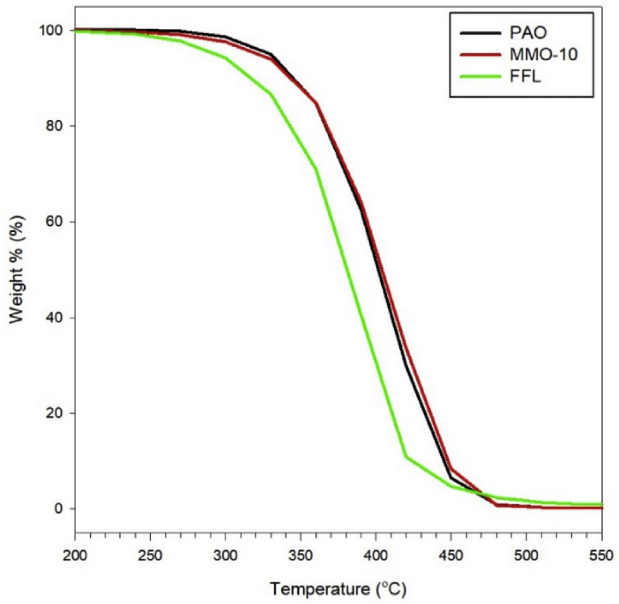


Fig. 3 – TGA of PAO, MMO-10 and FFL.

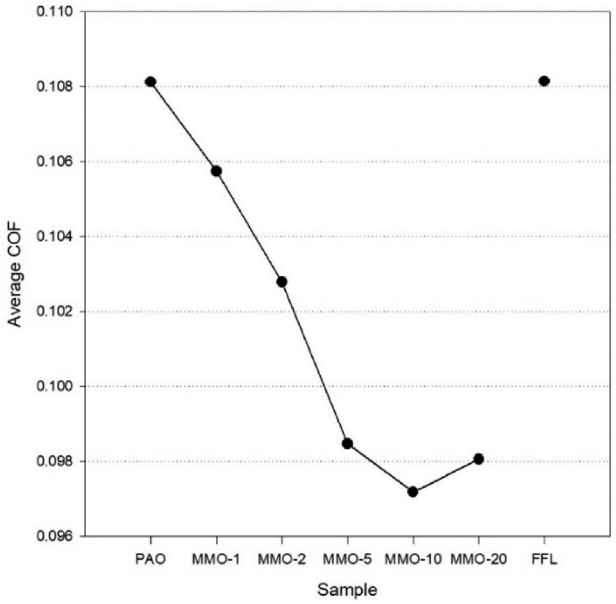


Fig. 5 – Average COF of PAO, FFL, and MMO blends.

the ability of MMO to dissipate heat may provide another valuable benefit for biolubricant application.

Friction characteristics

Fig. 5 shows the average COF of the PAO, FFL, and MMO blends over a test duration of 60 min. The COF is lower by 2.2–10.1% for blends containing up to 10 (v/v)% of MMO relative to that for PAO. The increase in blending percentage increases the ester content, thus providing more polar functional groups for the establishment of a stronger lubricating film due to the presence of higher amount of fatty acids available for close-packing against each other [63]. On the other hand, PAO has no polar functional groups; this prevented the hydrocarbons presence from being absorbed strongly on to the interacting surfaces. When the lubricating film is non-existence or very weak, increase of friction is expected since there is no molecular barrier to separate the rubbing surfaces, increasing the metal-to-metal contact.

However, there is no further improvement in the COF when the amount of MMO in the blend is increased to 20 (v/v)%. The COF is slightly higher for MMO-20 compared with that for MMO-10, which is likely due to corrosive wear, which overrides the effectiveness of the lubricating film, considering that MMO-20 has the highest acid value (13.688 mg KOH/g) among all of the lubricants. The higher acidity has caused the corrosion of metal, the break-off of the tribofilm (oxide layer) and the weakening of the molecular lubricating film, and these, collectively increase the COF. Interestingly, the average COF is similar for PAO and FFL; however, generally, the COF values are higher for these synthetic lubricants compared with those for the MMO blends. The lower COF indicate that MMO lubricated runs have less friction than PAO and FFL; indicating better lubricity.

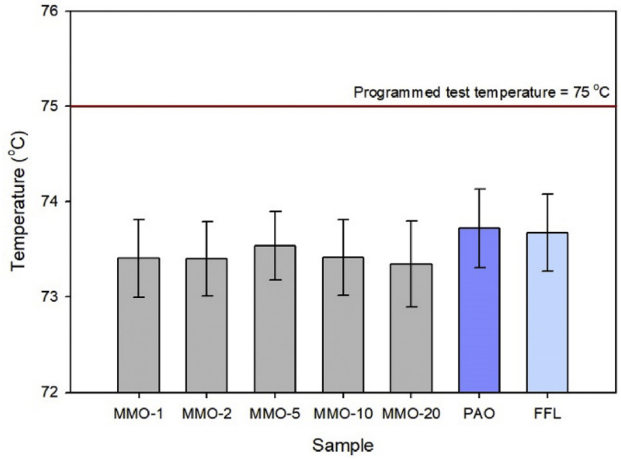


Fig. 4 – Average temperature measured during the stable friction phase.

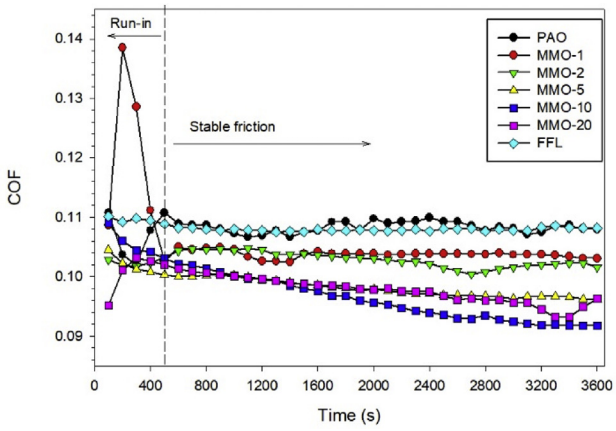


Fig. 6 – Variation of the COF with respect to time for the PAO, FFL, and MMO blends during the tribological test.

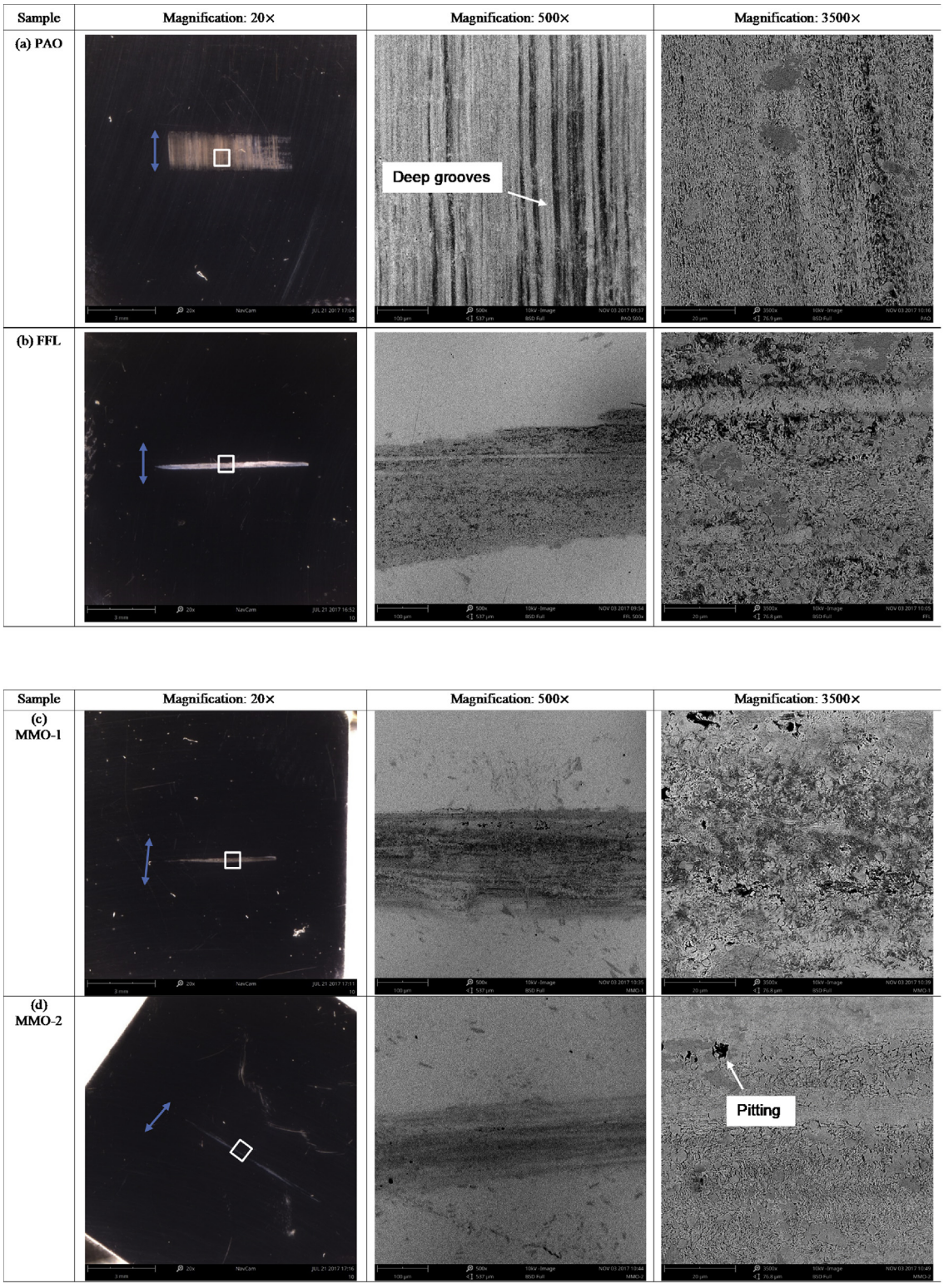


Fig. 7 – SEM images (magnification: 20×, 500×, and 3500×) of the carbon steel plate surfaces lubricated with (a) PAO, (b) FFL, (c) MMO-1, (d) MMO-2, (e) MMO-5, (f) MMO-10, and (g) MMO-20.

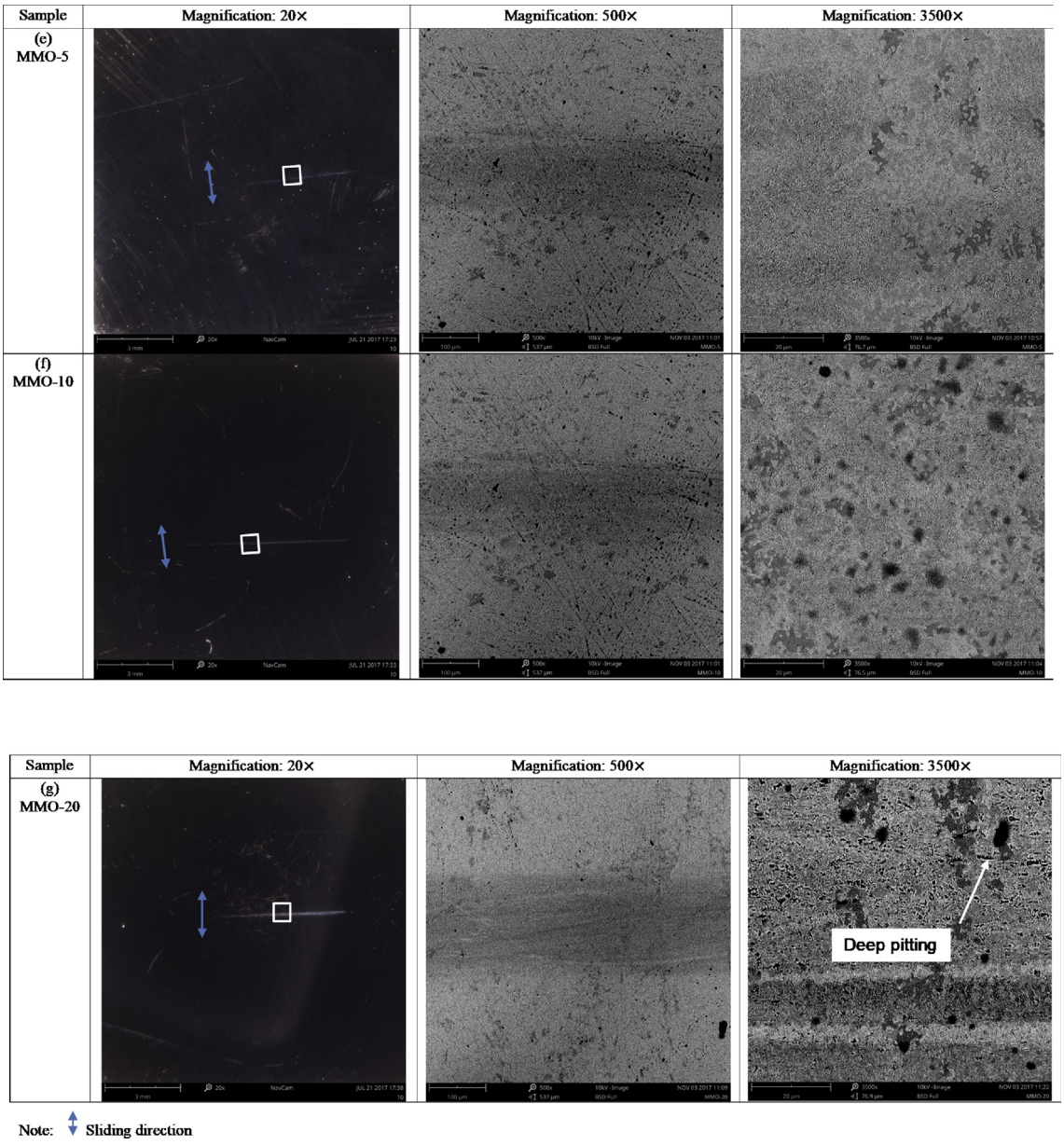


Fig. 7 – (*continued*).

Fig. 6 shows the variation of COF with respect to time for the PAO, FFL, and MMO blends during the tribological test. It can be observed that the COF stabilizes 6–8 min after the start of the tribological test (i.e., run-in phase). Mosarof et al. [64] also observed a run-in phase, though they used *Calophyllum inophyllum* and palm biodiesel blends in their experiments.

Table 7 – EDS elemental analysis of the carbon steel plate surfaces after tribological test.

Specimen	Element (Atomic %)								
	Fe	O	Cr	C	Co	N	Zn	P	Si
PAO	60.30	25.33	12.93	–	–	–	–	–	1.44
FFL	41.03	16.10	9.83	7.45	15.47	6.68	2.79	0.65	–
MMO-1	58.53	20.86	14.10	6.51	–	–	–	–	–
MMO-2	43.31	15.47	8.36	7.62	15.58	9.66	–	–	–
MMO-5	56.42	16.71	11.85	7.13		7.89	–	–	–
MMO-10	40.63	13.83	9.99	13.44	14.73	7.38	–	–	–
MMO-20	51.91	18.55	12.06	9.40		8.09	–	–	–

Note: Fe: Iron; O: Oxygen; Cr: Chromium; C: Carbon; Co: Cobalt; N: Nitrogen; Zn: Zinc; P: Phosphorus; Si: Silicon.

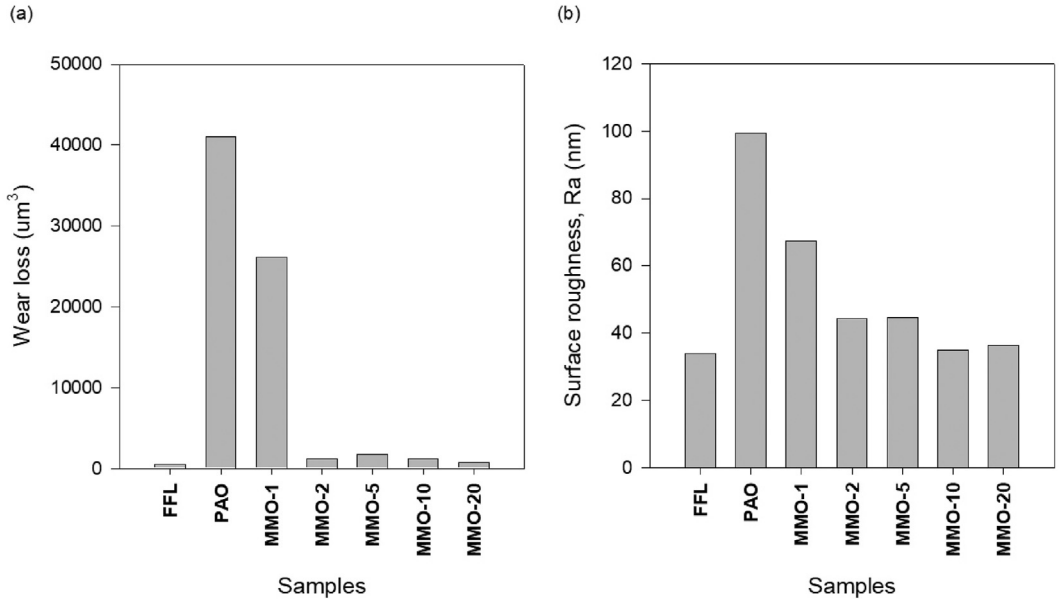


Fig. 8 – (a) Wear loss (volume) and (b) surface roughness of the lubricated surfaces.

They found that the duration of the run-in phase is dependent on the physical, chemical, and geometrical characteristics of the contact surfaces as well as the ability of the lubricant to provide wear protection. Commercial FFLs are typically formulated with friction improvers and dispersant in order to improve dispersion of the lubricants on the contact surfaces, as in this study, it is evident that the FFL demonstrates stable friction characteristics from the start of the tribological test. As mentioned earlier, ester-based lubricant and ester-added diesel also demonstrate the overall improvement of friction due to the adsorption of the polar functional groups on the mating surfaces and the close packing of the hydrocarbon chains, creating a molecular barrier during sliding [65,66]. Like any other lubricant, how readily and homogenously the ester groups disperse and adsorbed on the metal surfaces will too affect the run-in period [67]. Based on our results, the COF is lower for all MMO blends compared with those for PAO and FFL. In addition, it can be seen that the COF decreases with respect to time for the MMO-5, MMO-10, and MMO-20 blends, which is expected due to the presence of more functional groups for the creation of a thicker lubricating film, improving the lubricity.

Wear characteristics

Fig. 7 shows the SEM images of wear scar on the rubbed surfaces. It is apparent that there is severe scratching on the PAO-lubricated surface with a series of deep grooves parallel to the sliding direction, which is indicative of abrasive wear where materials are transferred by ploughing (Fig. 7a). The copper color of the wear scar indicates the formation of iron oxide at the rubbing interface. Iron oxide is a natural oxide layer that forms when the iron is exposed to air. It is the simplest form of wear protection but they are easily removed, even under a light load. Based on the results of the EDS elemental analysis (Table 7), the PAO-lubricated surface has the highest oxygen content among all of the tested surfaces confirming the formation of the iron oxide layer (tribofilm).

Wear is significantly reduced on the MMO-1-lubricated surface (Fig. 7c). The wear scar width is 0.23 mm, which is significantly lower compared with that for the PAO-lubricated surface (1.76 mm). Even at higher magnification (500×), it can be observed that only pits are visible and there are no grooves on the surface. Wear is further reduced for the surfaces lubricated with MMO blends containing higher concentrations of MMO. For instance, only shallow pits are visible on the MMO-5-lubricated surface (Fig. 7e) and these pits are less apparent on the MMO-10-lubricated surface (Fig. 7f). Based on the results of the EDS elemental analysis (Table 7), the MMO-10-lubricated surface has the lowest oxygen content but the highest carbon content among all of the tested surfaces, indicating that there is a large number of fatty acid chains present to form a lubricating film as well as lower oxidation rate. Adhesive wear is found to be the main mode of wear for the MMO-lubricated surfaces. Adhesion may first occur at the asperities that serve as contact sites and when these contact sites are sheared by sliding, detachment of fragments occurs. Some of these fragments may reattach to the same or to another surface, or may even form loose particles as sliding continues [68]. Further, multiple dark-bottom-pits that are corrosion-induced can be seen on MMO-20-lubricated surface (Fig. 7g) and it is likely that extensive surface-originated spalling will occur with prolonged exposure. Compared with PAO-lubricated surface, the wear scar width is reduced significantly on the FFL-lubricated surface (Fig. 7b). The FFL-lubricated surface has a few large pits compared with the PAO-lubricated surface, which has many small pits, considering that the COF is similar for both lubricants. This may be due to anti-scuffing properties of the FFL, which is a commercial synthetic lubricant. Typically, anti-scuff additives are non-metals such

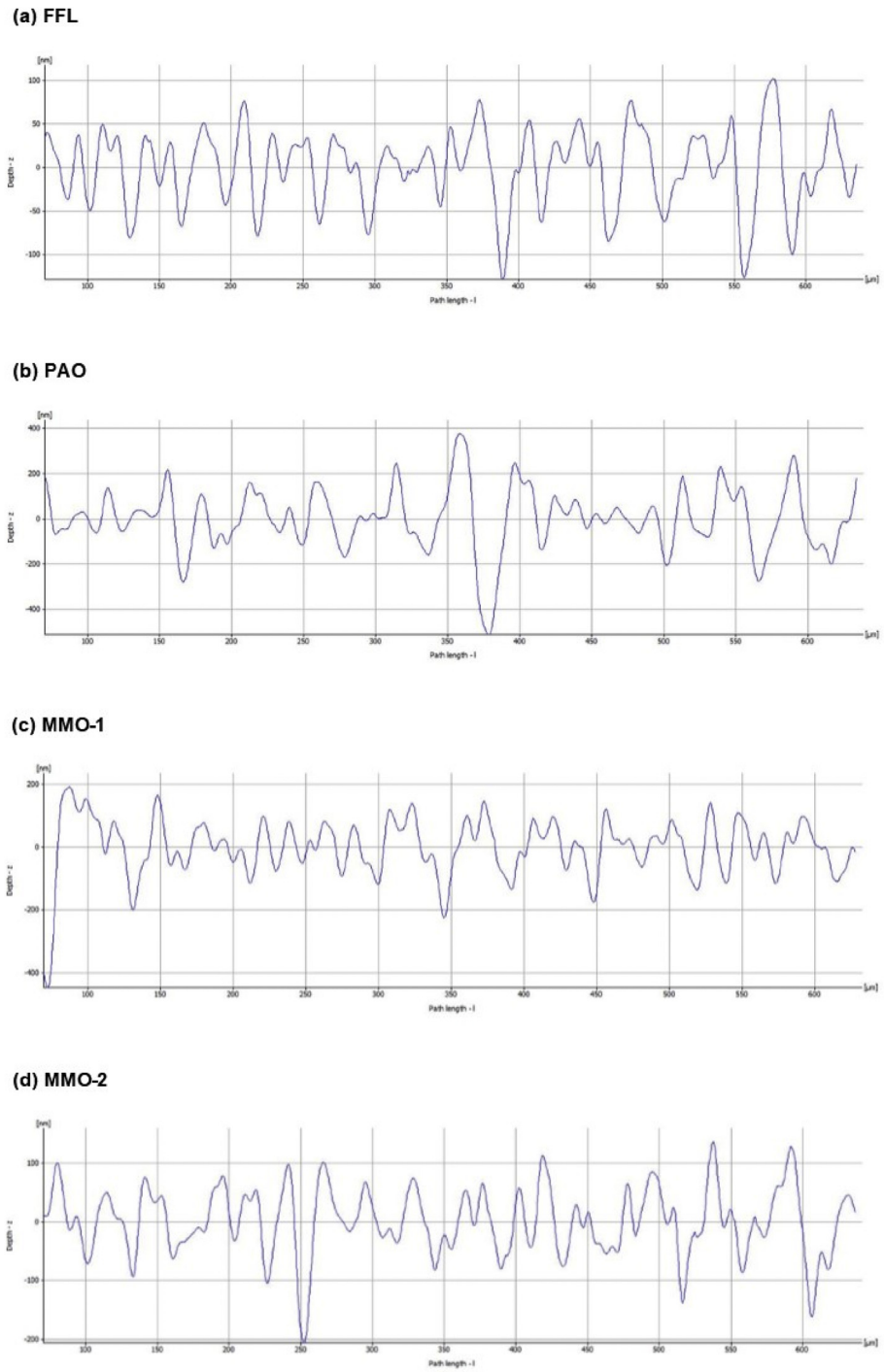


Fig. 9 – Profile curve of wear scar of (a) FFL, (b) PAO, (c) MMO-1, (b) MMO-2, (d) MMO-5, (e) MMO-10, and (f) MMO-20.

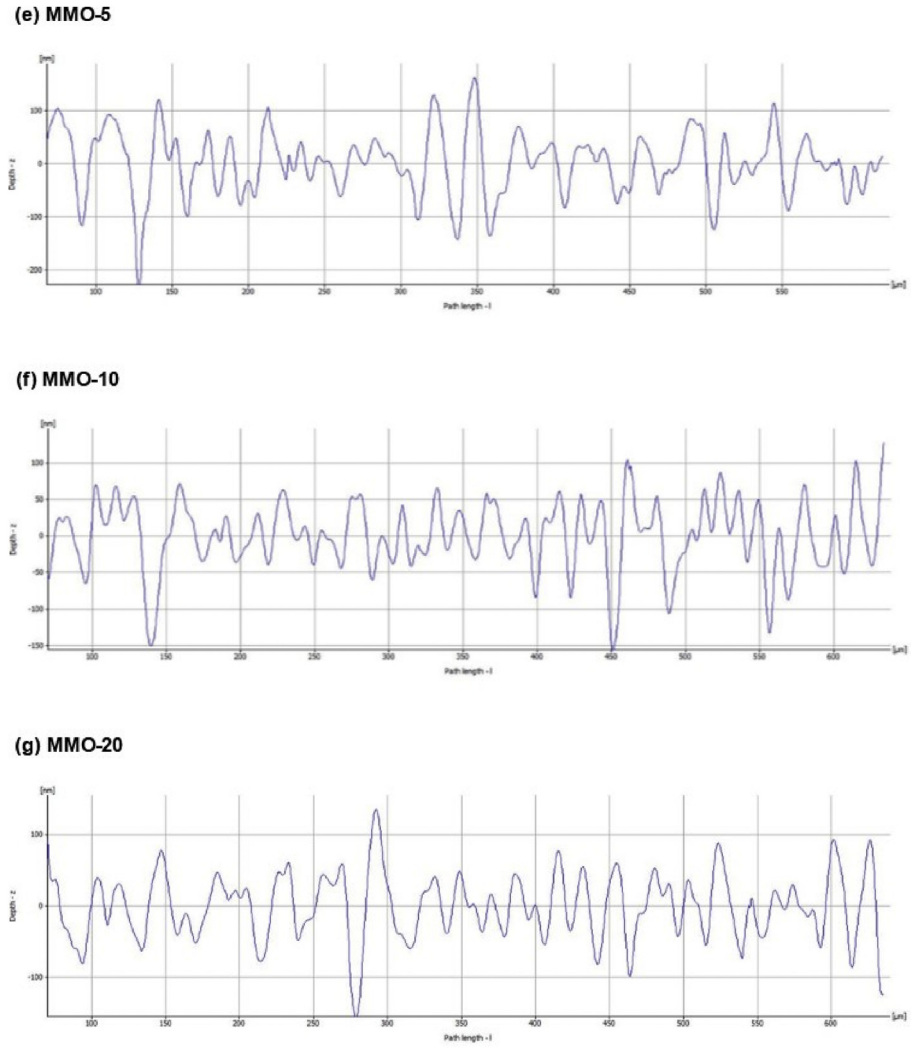


Fig. 9 – (*continued*).

as sulfur and phosphorus that reacted with the exposed metal surfaces to form a low shear strength tribofilm. This film helps prevent scuffing during sliding [69] where it is sacrificed to wear first. Based on the results of the EDS elemental analysis (Table 7), phosphorus is detected on the FFL-lubricated surface, suggesting that the FFL may contain an anti-scuffing additive.

Table 7 shows elemental composition of the lubricated surfaces after the tribological test. The presence of carbon and oxygen on the MMO blend-lubricated surfaces indicates that the polar long chain fatty acid compounds are physically adsorbed onto the oxide layers on the metal surfaces, forming a lubricating film that reduces friction and wear. Carbon is not detected on the PAO-lubricated surface, which suggests that there are no polar heads for metal surface attachment.

Wear loss and surface roughness

Significant reductions in wear loss and surface roughness were recorded for surfaces lubricated with MMO blends in comparison to PAO, refer Fig. 8. It is found that with the addition of 1% of MMO, wear loss and surface roughness are reduced 36.3 and 32.1% respectively, compared with PAO. When the MMO percentage increases to 10% (MMO-10), wear loss and surface roughness reductions of up to 97.0% and 64.8% are obtained in comparison to PAO. In a mechanical system, severe materials wear loss could lead to eventual system failure. Based on our experiment, the wear losses (volume) obtained for MMO lubricated runs (with more than 2% (v/v) blend) are comparable to the wear loss of the commercial FFL, which indicates that MMO possesses good anti-wear properties.

Fig. 9 shows the profiles of the wear scar on the rubbing surfaces parallel to the sliding direction. Generally, the depths of the wear scar on MMO-lubricated surfaces were not only less than that of the PAO-lubricated surface but the width of the valleys were also smaller in comparison to PAO lubricated surfaces. This was consistent with the SEM results in Fig. 7, where deep grooves were present on PAO-lubricated surfaces indicating abrasive wear. Meanwhile, on MMO-lubricated surfaces, no apparent grooves were detected.

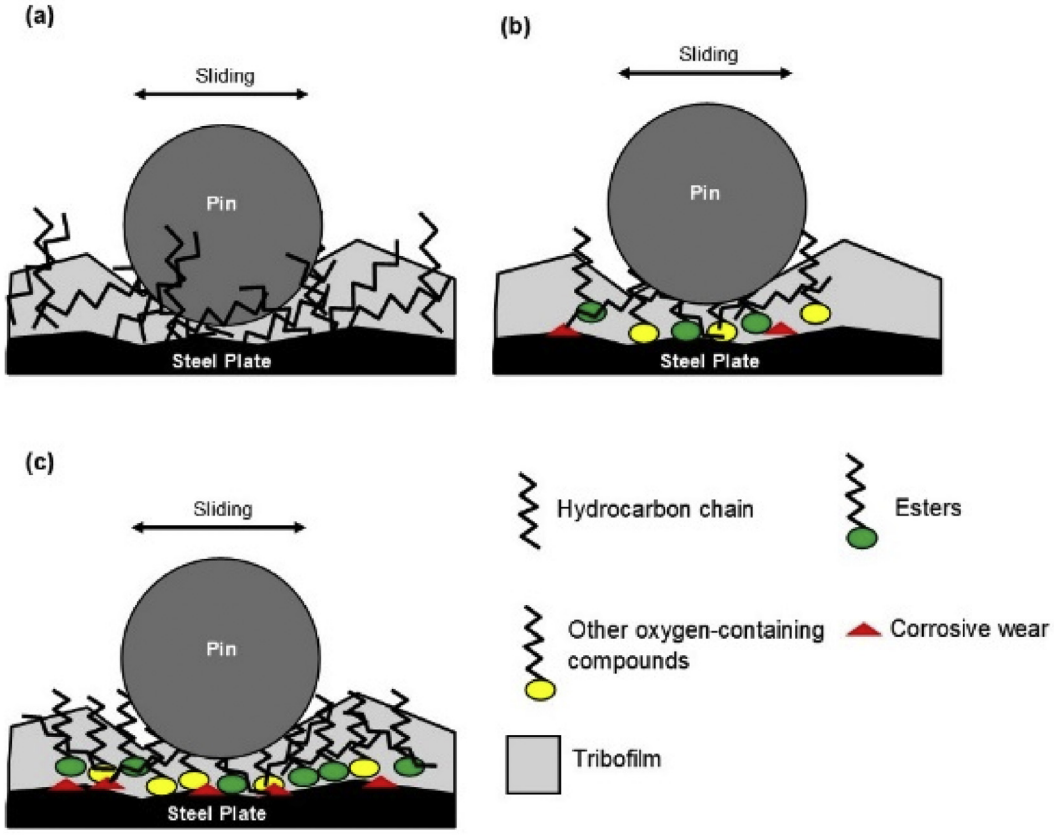


Fig. 10 – Schematic diagram of lubrication mechanisms for (a) PAO, (b) MMO and (c) higher concentration of MMO.

Lubrication mechanism

Based on the analysis performed, the lubricating mechanism of different samples can be summarized in Fig. 10. Both the tribofilm (oxide layer) and the absorption of the molecular lubricating film on to the mating surfaces contributed to the tribological characteristics of the samples. Only the oxide layer (tribofilm) was present on PAO-lubricated surface as carbon was not detected as shown by the EDS results. The absence of the polar functional groups from the PAO lubricant has prevented the formation of a thin lubricating film on the PAO-lubricated surfaces. Therefore, more direct metal-to-metal contact occurred (especially at the prominent asperities) when the surface is protected by only the oxide layer. Furthermore, more friction were generated without the presence of the molecular lubricating layer to assist the sliding of surfaces against each other. The continuous sliding has caused these asperities to scratch (plough) the other surface forming deep grooves as shown on the SEM image of PAO-lubricated surface compared with the MMO-lubricated surfaces.

On the other hand, carbon were detected on the MMO-lubricated surfaces signaling the presence of the hydrocarbon tails, the polar functional groups have enabled the absorption of esters and other oxygen-containing compounds onto the rubbing surfaces. These compounds assisted in the sliding of the surfaces against each other by the formation of a lubricating film. During sliding, the asperities on the two contacting surfaces in MMO-lubricated runs will attach (adhere) and detach in relative motion, causing metal to transfer from one surface to the other. Adhesive wear will cause the worn surfaces to be either rough or relatively smooth due to smearing and deformation of the materials [70]. In this study, the worn surfaces of MMO-lubricated runs were relatively smooth as evidenced in the SEM and surface roughness analysis compared with the PAO-lubricated surface. The increase of MMO concentration, increases the ester and oxygen-containing compounds for the formation of the lubricating film, but the acidity was found to increase as well causing tribo-corrosion on the rubbing surfaces. The acidic fluid has caused damage to the oxide layer and the breaking of the lubricating film where pits appeared on the exposed metal surfaces, particularly the MMO-20-lubricated surface, as shown in the SEM analysis.

Conclusions and future work

MMO has demonstrated great potential as a biolubricant, where the COF of MMO-lubricated runs are reduced up to 10.1%. Wear loss and surface roughness are significantly reduced in comparison to PAO-lubricated run. MMO addition also improves the heat dissipation of lubricant and MMO-10 was found to exhibit a slightly higher thermal stability than

FFL. Nonetheless, there are a few issues that need to be addressed such as the high acidity concern before MMO can be exploited for hydrogen-powered engine applications. High acidity will not only cause corrosion but also affects the storage stability of the lubricant. EP additive is used to protect the components in situations where the lubricating oil film thickness can no longer separate the interacting surfaces like in slow, highly loaded and geared applications. Currently, EP additives are produced from sulfurized fat, ester of chlorinated hydrocarbons, polymer esters, etc. which are not very environmental friendly. Future work will focus on the optimization of the esterification reaction to maximize the ester yield and to reduce the acidity. Furthermore, MMO performance under extreme pressure (EP) condition should also be investigated. Therefore, this study has demonstrated the potential and accompanied issues of utilizing microalgae for biolubricant application while the future work will centered on improving the quality and exploring additional usage for MMO.

Acknowledgment

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